Yamini Kandrekula

AI Engineer (LLM Systems | RAG | Multi-Agent Workflows)

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PROFESSIONAL SUMMARY

AI Engineer with 6+ years of experience building production-grade LLM systems, specializing in RAG pipelines, multi-agent architectures, and AI workflow automation. Proven track record of reducing manual workflows by up to 50% and improving response accuracy through retrieval optimization and prompt engineering. Experienced in deploying scalable AI systems on AWS serverless infrastructure.

SKILLS

AI / LLM Systems: LLM APIs (OpenAI, Claude), Prompt Engineering, RAG, Multi-Agent Systems, LangChain, AI Workflow Automation, Context Optimization.

AI Data & Retrieval: Embeddings, Vector Databases, Semantic Search, Chunking Strategies

Cloud & DevOps: AWS Lambda, S3, EventBridge, Firehose, CloudWatch, CI/CD

Backend & APIs: Node.js, Express, REST APIs, Webhooks, API Gateway

Frontend: React, Redux, Vue, HTML, CSS, SCSS

Architecture: Event-Driven Systems, Distributed Systems, Microservices

Data & Storage: MongoDB, Vector Databases (conceptual), Data Pipelines

Languages: JavaScript, TypeScript

Auth & Performance: JWT, RBAC, Caching, API Optimization.

KEY ACHIEVEMENTS

- Designed and implemented LLM evaluation and prompt optimization strategies, reducing hallucinations and improving response consistency in production systems.
- Built and deployed multi-agent AI systems automating complex workflows, reducing manual effort by over 50%
- Developed RAG-based pipelines integrating vector retrieval with LLMs, significantly improving contextual accuracy
- Architected event-driven distributed systems, improving scalability by 60% and reducing infrastructure costs by ~45%
- Optimized APIs and backend systems, reducing latency by 30-40% and improving system responsiveness.
- Recognized with multiple awards (Spot, GRACIAS) for high-impact engineering contributions.

WORK EXPERIENCE

Senior Software Engineer
Applaud, Hyderabad **November 2024 – Present**

- Designed and implemented a multi-agent AI system for job requisition automation, enabling dynamic data collection and structured content generation
- Built RAG pipelines using embeddings and vector search to enhance response accuracy and reduce hallucinations
- Engineered LLM-driven workflows integrating retrieval systems with downstream platforms (Workday)
- Implemented prompt optimization and context management strategies to improve output quality and domain alignment
- Integrated AI workflows with AWS serverless architecture (Lambda, EventBridge, S3) for scalable, asynchronous processing
- Designed low-latency APIs supporting real-time agent interactions, reducing response time by 30-40%
- Developed data pipelines for structured and unstructured data to support AI-driven workflows

Senior Software Engineer
LTIMindtree, Hyderabad **June 2021 – October 2024**

- Led full-stack MERN development for scalable enterprise apps.
- Reduced frontend load time by 35% using lazy loading & code splitting.
- Improved system maintainability, reducing production issues by 25%.
- Mentored developers and improved delivery velocity across team

Program Analyst
Cognizant, Chennai **December 2019 – June 2021**

- Developed MERN applications & scalable REST APIs for business workflows.
- Improved API performance by 25%+ via query optimization.
- Designed and developed REST APIs supporting high-throughput applications.
- Implemented JWT + RBAC, strengthening system security.

Software Engineer
Mojoreads, Vizag **March 2019 – September 2019**

- Developed React + Node.js applications with MongoDB.
- Improved UI responsiveness & resolved production issues.

PROJECTS

Agentic RAG System for Real-Time Market Intelligence (LLM + Multi-Agent)

- Built a multi-agent AI system that ingests real-time data (news, signals, APIs) and generates contextual insights.
- Designed a RAG architecture using embeddings and vector retrieval for accurate information grounding
- Implemented agent workflows for data ingestion → retrieval → reasoning → summarization
- Integrated external APIs for real-time data ingestion (news, market signals) to ensure up-to-date outputs
- Reduced hallucinations and improved response accuracy through prompt engineering, retrieval optimization, and context tuning
- Designed evaluation strategies for accuracy, relevance, and consistency to ensure reliable AI outputs

Serverless Event Processing System (AWS)

- Designed event-driven pipelines using Lambda + EventBridge
- Reduced infra cost by ~45% with serverless architecture
- Built scalable ingestion pipeline using S3 + Firehose
- Implemented rate limiting and caching

EDUCATION

Bachelor of Technology

Anil Neerukonda Institute of Technology & Sciences, Vizag **Graduated: 2019**